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```
import numpy as np
import pickle
import cv2
from os import listdir
from sklearn.preprocessing import LabelBinarizer
from keras.models import Sequential
from keras.layers.normalization import BatchNormalization
from keras.layers.convolutional import Conv2D
from keras.layers.convolutional import MaxPooling2D
from keras.layers.core import Activation, Flatten, Dropout, Dense
from keras import backend as K
from keras.preprocessing.image import ImageDataGenerator
from keras.optimizers import Adam
from keras.preprocessing import image
from keras.preprocessing.image import img_to_array
from sklearn.preprocessing import MultiLabelBinarizer
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
```

```
Using TensorFlow backend.
/opt/conda/lib/python3.6/site-packages/tensorflow/python/framework/dtypes.py:516: FutureWarning: Passing (type, 1) or '1type
  _np_qint8 = np.dtype([("qint8", np.int8, 1)])
/opt/conda/lib/python3.6/site-packages/tensorflow/python/framework/dtypes.py:517: FutureWarning: Passing (type, 1) or '1type
  _np_quint8 = np.dtype([("quint8", np.uint8, 1)])
/opt/conda/lib/python3.6/site-packages/tensorflow/python/framework/dtypes.py:518: FutureWarning: Passing (type, 1) or '1type
  _np_qint16 = np.dtype([("qint16", np.int16, 1)])
/opt/conda/lib/python3.6/site-packages/tensorflow/python/framework/dtypes.py:519: FutureWarning: Passing (type, 1) or '1type
  _np_quint16 = np.dtype([("quint16", np.uint16, 1)])
/opt/conda/lib/python3.6/site-packages/tensorflow/python/framework/dtypes.py:520: FutureWarning: Passing (type, 1) or '1type
  _np_qint32 = np.dtype([("qint32", np.int32, 1)])
/opt/conda/lib/python3.6/site-packages/tensorflow/python/framework/dtypes.py:525: FutureWarning: Passing (type, 1) or '1type
  np_resource = np.dtype([("resource", np.ubyte, 1)])
/opt/conda/lib/python3.6/site-packages/tensorboard/compat/tensorflow_stub/dtypes.py:541: FutureWarning: Passing (type, 1) or
  _np_qint8 = np.dtype([("qint8", np.int8, 1)])
/opt/conda/lib/python3.6/site-packages/tensorboard/compat/tensorflow_stub/dtypes.py:542: FutureWarning: Passing (type, 1) or
  _np_quint8 = np.dtype([("quint8", np.uint8, 1)])
/opt/conda/lib/python3.6/site-packages/tensorboard/compat/tensorflow_stub/dtypes.py:543: FutureWarning: Passing (type, 1) or
  _np_qint16 = np.dtype([("qint16", np.int16, 1)])
/opt/conda/lib/python3.6/site-packages/tensorboard/compat/tensorflow_stub/dtypes.py:544: FutureWarning: Passing (type, 1) or
  _np_quint16 = np.dtype([("quint16", np.uint16, 1)])
/opt/conda/lib/python3.6/site-packages/tensorboard/compat/tensorflow_stub/dtypes.py:545: FutureWarning: Passing (type, 1) or
  _np_qint32 = np.dtype([("qint32", np.int32, 1)])
/opt/conda/lib/python3.6/site-packages/tensorboard/compat/tensorflow_stub/dtypes.py:550: FutureWarning: Passing (type, 1) or
  np_resource = np.dtype([("resource", np.ubyte, 1)])
```

```
EPOCHS = 25
INIT_LR = 1e-3
BS = 32
default_image_size = tuple((256, 256))
image_size = 0
directory_root = 'C:\\Users\\Dell\\Downloads\\archive (1)\\PlantVillage'
width=256
height=256
depth=3
```

Function to convert images to array

```
def convert_image_to_array(image_dir):
    try:
        image = cv2.imread(image_dir)
        if image is not None :
            image = cv2.resize(image, default_image_size)
            return img_to_array(image)
        else :
            return np.array([])
    except Exception as e:
        print(f"Error : {e}")
        return None
```

Fetch images from directory

```
image_list, label_list = [], []
try:
```

```
        print("[INFO] Loading images ...")
        root_dir = listdir(directory_root)
        for directory in root_dir :
            # remove .DS_Store from list
            if directory == ".DS_Store" :
                root_dir.remove(directory)

        for plant_folder in root_dir :
            plant_disease_folder_list = listdir(f"{directory_root}/{plant_folder}")

            for disease_folder in plant_disease_folder_list :
                # remove .DS_Store from list
                if disease_folder == ".DS_Store" :
                    plant_disease_folder_list.remove(disease_folder)

            for plant_disease_folder in plant_disease_folder_list:
                print(f"[INFO] Processing {plant_disease_folder} ...")
                plant_disease_image_list = listdir(f"{directory_root}/{plant_folder}/{plant_disease_folder}/")

                for single_plant_disease_image in plant_disease_image_list :
                    if single_plant_disease_image == ".DS_Store" :
                        plant_disease_image_list.remove(single_plant_disease_image)

                for image in plant_disease_image_list[:200]:
                    image_directory = f"{directory_root}/{plant_folder}/{plant_disease_folder}/{image}"
                    if image_directory.endswith(".jpg") == True or image_directory.endswith(".JPG") == True:
                        image_list.append(convert_image_to_array(image_directory))
                        label_list.append(plant_disease_folder)
        print("[INFO] Image loading completed")
    except Exception as e:
        print(f"Error : {e}")
```

```
[INFO] Loading images ...
[INFO] Processing Tomato_Septoria_leaf_spot ...
[INFO] Processing Tomato_Late_blight ...
[INFO] Processing Potato___Late_blight ...
[INFO] Processing Tomato_healthy ...
[INFO] Processing Pepper__bell___healthy ...
[INFO] Processing Tomato_Spider_mites_Two_spotted_spider_mite ...
[INFO] Processing Tomato_Bacterial_spot ...
[INFO] Processing Tomato_Early_blight ...
[INFO] Processing Tomato__Tomato_YellowLeaf__Curl_Virus ...
[INFO] Processing Tomato__Tomato_mosaic_virus ...
[INFO] Processing Tomato__Target_Spot ...
[INFO] Processing Potato___Early_blight ...
[INFO] Processing Pepper__bell___Bacterial_spot ...
[INFO] Processing Tomato_Leaf_Mold ...
[INFO] Processing Potato___healthy ...
[INFO] Image loading completed
```

Get Size of Processed Image

```
image_size = len(image_list)
```

Transform Image Labels uisng Scikit LabelBinarizer

```
label_binarizer = LabelBinarizer()
image_labels = label_binarizer.fit_transform(label_list)
pickle.dump(label_binarizer,open('label_transform.pkl', 'wb'))
n_classes = len(label_binarizer.classes_)
```

Print the classes

```
print(label_binarizer.classes_)
```

```
['Pepper__bell___Bacterial_spot' 'Pepper__bell___healthy'
 'Potato___Early_blight' 'Potato___Late_blight' 'Potato___healthy'
 'Tomato_Bacterial_spot' 'Tomato_Early_blight' 'Tomato_Late_blight'
 'Tomato_Leaf_Mold' 'Tomato_Septoria_leaf_spot'
 'Tomato_Spider_mites_Two_spotted_spider_mite' 'Tomato__Target_Spot'
 'Tomato__Tomato_YellowLeaf__Curl_Virus' 'Tomato__Tomato_mosaic_virus'
 'Tomato_healthy']
```

```
np_image_list = np.array(image_list, dtype=np.float16) / 225.0
```

```
print("[INFO] Spliting data to train, test")
x_train, x_test, y_train, y_test = train_test_split(np_image_list, image_labels, test_size=0.2, random_state = 42)
```

```
[INFO] Spliting data to train, test
```

```
aug = ImageDataGenerator(
    rotation_range=25, width_shift_range=0.1,
    height_shift_range=0.1, shear_range=0.2,
    zoom_range=0.2,horizontal_flip=True,
    fill_mode="nearest")
```

```
model = Sequential()
inputShape = (height, width, depth)
chanDim = -1
if K.image_data_format() == "channels_first":
    inputShape = (depth, height, width)
    chanDim = 1
model.add(Conv2D(32, (3, 3), padding="same",input_shape=inputShape))
model.add(Activation("relu"))
model.add(BatchNormalization(axis=chanDim))
model.add(MaxPooling2D(pool_size=(3, 3)))
model.add(Dropout(0.25))
model.add(Conv2D(64, (3, 3), padding="same"))
model.add(Activation("relu"))
model.add(BatchNormalization(axis=chanDim))
model.add(Conv2D(64, (3, 3), padding="same"))
model.add(Activation("relu"))
model.add(BatchNormalization(axis=chanDim))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Dropout(0.25))
model.add(Conv2D(128, (3, 3), padding="same"))
model.add(Activation("relu"))
model.add(BatchNormalization(axis=chanDim))
model.add(Conv2D(128, (3, 3), padding="same"))
model.add(Activation("relu"))
model.add(BatchNormalization(axis=chanDim))
model.add(MaxPooling2D(pool_size=(2, 2)))
model.add(Dropout(0.25))
model.add(Flatten())
model.add(Dense(1024))
model.add(Activation("relu"))
model.add(BatchNormalization())
model.add(Dropout(0.5))
model.add(Dense(n_classes))
model.add(Activation("softmax"))
```

Model Summary

```
model.summary()
```

```
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv2d_1 (Conv2D)            (None, 256, 256, 32)      896
_________________________________________________________________
activation_1 (Activation)    (None, 256, 256, 32)      0
_________________________________________________________________
batch_normalization_1 (Batch (None, 256, 256, 32)      128
_________________________________________________________________
max_pooling2d_1 (MaxPooling2 (None, 85, 85, 32)        0
_________________________________________________________________
dropout_1 (Dropout)          (None, 85, 85, 32)        0
_________________________________________________________________
conv2d_2 (Conv2D)            (None, 85, 85, 64)        18496
_________________________________________________________________
activation_2 (Activation)    (None, 85, 85, 64)        0
_________________________________________________________________
batch_normalization_2 (Batch (None, 85, 85, 64)        256
_________________________________________________________________
conv2d_3 (Conv2D)            (None, 85, 85, 64)        36928
_________________________________________________________________
activation_3 (Activation)    (None, 85, 85, 64)        0
_________________________________________________________________
batch_normalization_3 (Batch (None, 85, 85, 64)        256
_________________________________________________________________
max_pooling2d_2 (MaxPooling2 (None, 42, 42, 64)        0
_________________________________________________________________
dropout_2 (Dropout)          (None, 42, 42, 64)        0
_________________________________________________________________
conv2d_4 (Conv2D)            (None, 42, 42, 128)       73856
_________________________________________________________________
activation_4 (Activation)    (None, 42, 42, 128)       0
_________________________________________________________________
batch_normalization_4 (Batch (None, 42, 42, 128)       512
_________________________________________________________________
conv2d_5 (Conv2D)            (None, 42, 42, 128)       147584
_________________________________________________________________
```

```
activation_5 (Activation)    (None, 42, 42, 128)       0
_________________________________________________________________
batch_normalization_5 (Batch (None, 42, 42, 128)       512
_________________________________________________________________
max_pooling2d_3 (MaxPooling2 (None, 21, 21, 128)       0
_________________________________________________________________
dropout_3 (Dropout)          (None, 21, 21, 128)       0
_________________________________________________________________
flatten_1 (Flatten)          (None, 56448)             0
_________________________________________________________________
dense_1 (Dense)              (None, 1024)              57803776
_________________________________________________________________
activation_6 (Activation)    (None, 1024)              0
_________________________________________________________________
batch_normalization_6 (Batch (None, 1024)              4096
_________________________________________________________________
dropout_4 (Dropout)          (None, 1024)              0
_________________________________________________________________
dense_2 (Dense)              (None, 15)                15375
_________________________________________________________________
```

```
opt = Adam(lr=INIT_LR, decay=INIT_LR / EPOCHS)
# distribution
model.compile(loss="binary_crossentropy", optimizer=opt,metrics=["accuracy"])
# train the network
print("[INFO] training network...")
```

```
[INFO] training network...
```

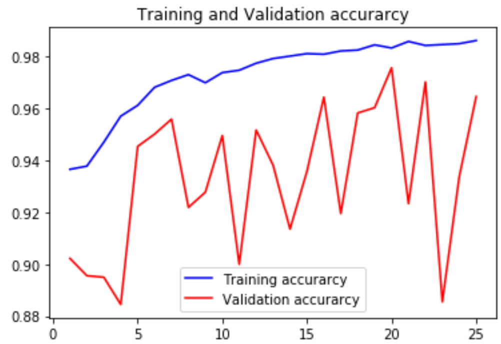
```
history = model.fit_generator(
    aug.flow(x_train, y_train, batch_size=BS),
    validation_data=(x_test, y_test),
    steps_per_epoch=len(x_train) // BS,
    epochs=EPOCHS, verbose=1
    )
```

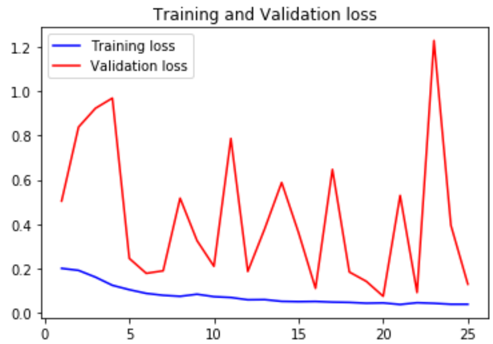
```
Epoch 1/25
73/73 [==============================] - 43s 583ms/step - loss: 0.2012 - acc: 0.9365 - val_loss: 0.5047 - val_acc: 0.9022
Epoch 2/25
73/73 [==============================] - 35s 479ms/step - loss: 0.1920 - acc: 0.9378 - val_loss: 0.8381 - val_acc: 0.8955
Epoch 3/25
73/73 [==============================] - 36s 491ms/step - loss: 0.1617 - acc: 0.9469 - val_loss: 0.9227 - val_acc: 0.8950
Epoch 4/25
73/73 [==============================] - 36s 490ms/step - loss: 0.1249 - acc: 0.9569 - val_loss: 0.9686 - val_acc: 0.8845
Epoch 5/25
73/73 [==============================] - 34s 470ms/step - loss: 0.1052 - acc: 0.9611 - val_loss: 0.2461 - val_acc: 0.9453
Epoch 6/25
73/73 [==============================] - 36s 487ms/step - loss: 0.0884 - acc: 0.9680 - val_loss: 0.1789 - val_acc: 0.9500
Epoch 7/25
73/73 [==============================] - 34s 468ms/step - loss: 0.0798 - acc: 0.9707 - val_loss: 0.1902 - val_acc: 0.9558
Epoch 8/25
73/73 [==============================] - 35s 483ms/step - loss: 0.0752 - acc: 0.9729 - val_loss: 0.5176 - val_acc: 0.9218
Epoch 9/25
73/73 [==============================] - 35s 476ms/step - loss: 0.0848 - acc: 0.9698 - val_loss: 0.3263 - val_acc: 0.9277
Epoch 10/25
73/73 [==============================] - 35s 482ms/step - loss: 0.0737 - acc: 0.9736 - val_loss: 0.2105 - val_acc: 0.9495
Epoch 11/25
73/73 [==============================] - 35s 481ms/step - loss: 0.0699 - acc: 0.9746 - val_loss: 0.7867 - val_acc: 0.8999
Epoch 12/25
73/73 [==============================] - 35s 473ms/step - loss: 0.0598 - acc: 0.9772 - val_loss: 0.1873 - val_acc: 0.9516
Epoch 13/25
73/73 [==============================] - 35s 478ms/step - loss: 0.0604 - acc: 0.9791 - val_loss: 0.3801 - val_acc: 0.9381
Epoch 14/25
73/73 [==============================] - 34s 468ms/step - loss: 0.0526 - acc: 0.9800 - val_loss: 0.5882 - val_acc: 0.9135
Epoch 15/25
73/73 [==============================] - 36s 487ms/step - loss: 0.0512 - acc: 0.9810 - val_loss: 0.3619 - val_acc: 0.9357
Epoch 16/25
73/73 [==============================] - 35s 482ms/step - loss: 0.0520 - acc: 0.9808 - val_loss: 0.1112 - val_acc: 0.9642
Epoch 17/25
73/73 [==============================] - 34s 472ms/step - loss: 0.0491 - acc: 0.9820 - val_loss: 0.6471 - val_acc: 0.9195
Epoch 18/25
73/73 [==============================] - 35s 484ms/step - loss: 0.0478 - acc: 0.9824 - val_loss: 0.1848 - val_acc: 0.9582
Epoch 19/25
73/73 [==============================] - 34s 469ms/step - loss: 0.0443 - acc: 0.9843 - val_loss: 0.1428 - val_acc: 0.9602
Epoch 20/25
73/73 [==============================] - 35s 483ms/step - loss: 0.0455 - acc: 0.9832 - val_loss: 0.0754 - val_acc: 0.9755
Epoch 21/25
73/73 [==============================] - 35s 476ms/step - loss: 0.0384 - acc: 0.9857 - val_loss: 0.5299 - val_acc: 0.9233
Epoch 22/25
73/73 [==============================] - 35s 481ms/step - loss: 0.0463 - acc: 0.9842 - val_loss: 0.0925 - val_acc: 0.9701
Epoch 23/25
73/73 [==============================] - 35s 481ms/step - loss: 0.0440 - acc: 0.9844 - val_loss: 1.2281 - val_acc: 0.8855
Epoch 24/25
73/73 [==============================] - 34s 472ms/step - loss: 0.0396 - acc: 0.9847 - val_loss: 0.3955 - val_acc: 0.9337
Epoch 25/25
73/73 [==============================] - 35s 483ms/step - loss: 0.0394 - acc: 0.9860 - val_loss: 0.1300 - val_acc: 0.9645
```

Plot the train and val curve

```
acc = history.history['acc']
val_acc = history.history['val_acc']
loss = history.history['loss']
val_loss = history.history['val_loss']
epochs = range(1, len(acc) + 1)
#Train and validation accuracy
plt.plot(epochs, acc, 'b', label='Training accurarcy')
plt.plot(epochs, val_acc, 'r', label='Validation accurarcy')
plt.title('Training and Validation accurarcy')
plt.legend()

plt.figure()
#Train and validation loss
plt.plot(epochs, loss, 'b', label='Training loss')
plt.plot(epochs, val_loss, 'r', label='Validation loss')
plt.title('Training and Validation loss')
plt.legend()
plt.show()
```





Model Accuracy

```
print("[INFO] Calculating model accuracy")
scores = model.evaluate(x_test, y_test)
print(f"Test Accuracy: {scores[1]*100}")
```

```
[INFO] Calculating model accuracy
591/591 [==============================] - 1s 2ms/step
Test Accuracy: 96.44670223221561
```

Save model using Pickle

```
# save the model to disk
print("[INFO] Saving model...")
pickle.dump(model,open('cnn_model.pkl', 'wb'))
```

```
[INFO] Saving model...
```